LEX PROGRAM TO IDENTIFY ASSIGNMENT OPERATORS

1. AIM

To write and execute a Lex program that identifies and prints the assignment operators used in a given C program, namely =, +=, -=, *=, and /=.

2. PSEUDOCODE

1. Start the program.
2. Enable noyywrap so scanning ends at end-of-file.
3. Define patterns for +=, -=, *=, /=, and =.
4. When an assignment operator is matched, print it.
5. Ignore whitespace and all other characters.
6. Read input until end-of-file is reached.
7. Stop the program.

3. LEX PROGRAM

```
%option noyywrap

%{
#include <stdio.h>
%}

%%
"+="    { printf("Assignment operator: +=\n"); }
"-="    { printf("Assignment operator: -=\n"); }
"*="    { printf("Assignment operator: *=\n"); }
"/="    { printf("Assignment operator: /=\n"); }
"="     { printf("Assignment operator: =\n"); }
[ \t\n]+ ;
. ;
%%

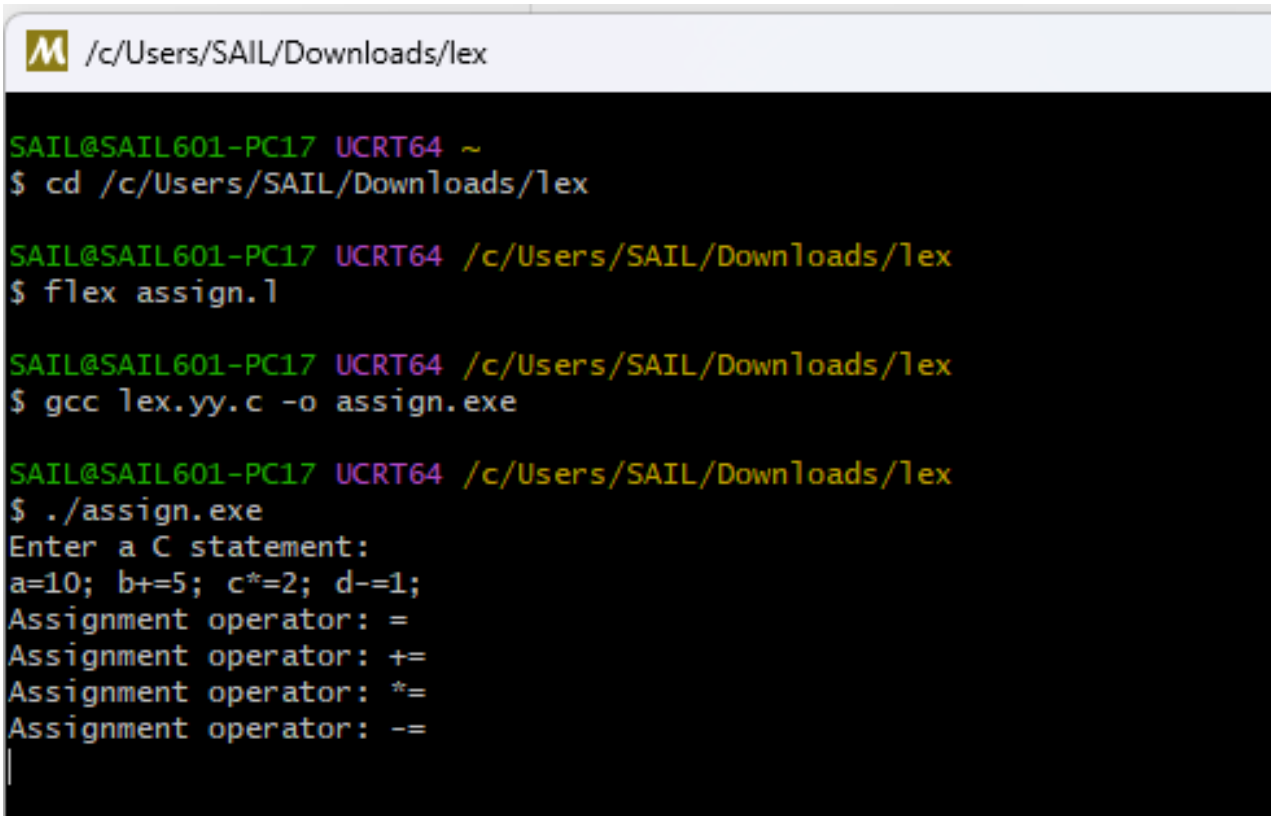
int main(void)
{
    printf("Enter a C statement:\n");
    yylex();
    return 0;
}
```

4. SAMPLE INPUT

```
a=10; b+=5; c*=2; d-=1;
```

5. OUTPUT IMAGE

The following screenshot shows the program executed in the MSYS2 UCRT64 terminal.



6. RESULT

Thus, the Lex program to identify and print assignment operators in a given C statement was implemented and executed. For the sample input, the program identifies =, +=, *=, and -= in the order in which they occur.